Standard Operating Procedure for Resolving BGP Route Churn, OSPF Adjacency Loss, and ISIS LSP Flooding on Cisco PE Node KR-PE-15 at KR Puram Access

Node: KR-PE-15 | **Vendor:** Cisco | **Severity:** Warning | **Alarm Code:** ALARM_4745

Description

The failure domain of Routing Protocol Flap on Cisco PE Node KR-PE-15 at KR Puram Access is a consequence of the complex interplay between BGP, OSPF, and ISIS protocols. BGP route churn occurs when there are frequent advertisements of route updates, leading to oscillations in BGP routing tables, while OSPF adjacency loss results from issues with OSPF neighbor establishment or adjacency maintenance. Meanwhile, ISIS LSP flooding is triggered by excessive flooding of IS-IS link-state advertisements (LSAs). These events can be caused by various factors, including misconfigured routing policies, inconsistencies in BGP AS_PATH or IGP metrics, or hardware malfunctions. The combination of these protocol-specific issues can lead to a cascading effect, resulting in a Routing Protocol Flap. To address this failure domain, it is essential to investigate and rectify the root causes of the BGP route churn, OSPF adjacency loss, and ISIS LSP flooding by analyzing the control plane operations and optimizing the routing policies, while ensuring that any changes do not compromise the stability or security of the network.

Pre-requisite Checks

- Ensure that the KR Puram Access site is accessible and that there are no power or environmental issues.

- Verify the status of the redundant links and ensure that they are functioning correctly.

- Check the BGP AS_PATH and IGP metrics for inconsistencies and adjust them as necessary.

Potential Root Causes

Cause	Probability
Misconfigured routing policies	High
Inconsistent BGP AS_PATH or IGP metrics	Medium
Hardware malfunction	Low

Diagnostic Commands & Expected Outputs

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Command: show ip bgp summary, Expected Output: BGP table version is 4, main routing table version is 4 BGP scan timer is 60 secs Status codes: s-suppressed, d-damped, h-hold, *-valid, >-best, i-internal, r-RIB-failure, S-stale Origin codes: i - IGP, e - EGP, ? - incomplete
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Network Next Hop Metric LocPrf Weight Path, Description: Isolate the BGP routing table and check for any anomalies or inconsistencies in the BGP AS PATH or IGP metrics.

Command: show ip ospf database, Expected Output: OSPF Router with ID (192.168.1.1) (Process ID 1) Router Link States (Area 0) Link ID ADV Router Age Seq# Fragment ID Metric 192.168.1.1 192.168.1.1 126 0x80000001 0x50000000 2 , Description: Examine the OSPF database and verify the OSPF neighbor relationships and adjacency status.

Command: show isis database, Expected Output: IS-IS Level-1 Link-State Database: LS age: 1235 Sequence number: 0x1000(0x00000001) Checksum: 0x4F45 Length: 96 IS IS Hello: 1 seconds, 3 attempts, Max NSDPs: 4 IS IS Metric: 10 IS IS Path: 1 IS IS Metric: 10, Description: Analyze the ISIS database and check for any issues with ISIS LSP flooding or excessive flooding of IS-IS LSA packets.
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Resolution Steps

Step 1: Step: Verify BGP routing policies, Command: show ip bgp filter-list, Description: Check the BGP routing policies and verify that they are correctly configured and not causing any issues with BGP route churn.

Step 2: Step: Adjust BGP AS_PATH and IGP metrics, Command: router bgp 100 neighbor 192.168.1.2 remote-as 200 neighbor 192.168.1.2 update-source Loopback0 neighbor 192.168.1.2 next-hop-self neighbor 192.168.1.2 soft-reconfiguration inbound, Description: Adjust the BGP AS_PATH and IGP metrics as necessary to resolve the BGP route churn and OSPF adjacency loss issues.

Step 3: Step: Optimize ISIS LSP flooding, Command: router isis net 47.0000.0000.0000.0000.00 router isis net 47.0000.0000.0000.0000.00 isis max-lsp-lifetime 1200 isis max-lsp-refresh 300, Description: Optimize the ISIS LSP flooding by adjusting the max-lsp-lifetime and max-lsp-refresh timers to resolve the ISIS LSP flooding issue.

Post-Check Verification

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Command: show ip bgp summary, Expected Output: BGP table version is 4, main routing table version is 4 BGP scan timer is 60 secs Status codes: s-suppressed, d-damped, h-hold, *-valid, >-best, i-internal, r-RIB-failure, S-stale Origin codes: i - IGP, e - EGP, ? - incomplete Network Next Hop Metric LocPrf Weight Path, Description: Verify the BGP routing table and check for any anomalies or inconsistencies in the BGP AS_PATH or IGP metrics.

Command: show ip ospf database, Expected Output: OSPF Router with ID (192.168.1.1) (Process ID 1) Router Link States (Area 0) Link ID ADV Router Age Seq# Fragment ID Metric 192.168.1.1 192.168.1.1 126 0x80000001 0x50000000 2 , Description: Examine the OSPF database and verify the OSPF neighbor relationships and adjacency status.

Command: show isis database, Expected Output: IS-IS Level-1 Link-State Database: LS age: 1235 Sequence number: 0x1000(0x00000001) Checksum: 0x4F45 Length: 96 IS IS Hello: 1 seconds, 3 attempts, Max NSDPs: 4 IS IS Metric: 10 IS IS Path: 1 IS IS Metric: 10, Description: Analyze the ISIS database and check for any issues with ISIS LSP flooding or excessive flooding of IS-IS LSA packets.
```

Roll-back Procedures

- Step: Revert BGP routing policies, Command: show ip bgp filter-list | diff -u, Description: Revert the BGP routing policies to their original state if the fix fails.

- Step: Revert BGP AS_PATH and IGP metrics, Command: router bgp 100 neighbor 192.168.1.2 remote-as 200 neighbor 192.168.1.2 update-source Loopback0 neighbor 192.168.1.2 next-hop-self

neighbor 192.168.1.2 soft-reconfiguration inbound | diff -u, Description: Revert the BGP AS_PATH and IGP metrics to their original state if the fix fails.

- Step: Revert ISIS LSP flooding, Command: router isis net 47.0000.0000.0000.0000.00 router isis net 47.0000.0000.0000.0000.00 isis max-lsp-lifetime 1200 isis max-lsp-refresh 300 | diff -u, Description: Revert the ISIS LSP flooding to its original state if the fix fails.